



Georgia Tech
College of
Engineering

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 20 – Generalized Hooke's law

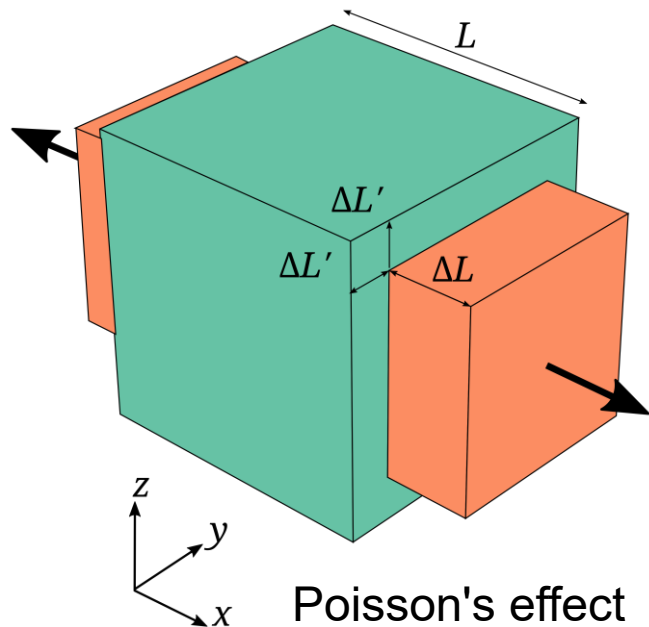
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Georgia Institute of Technology

Apr. 13, 2026

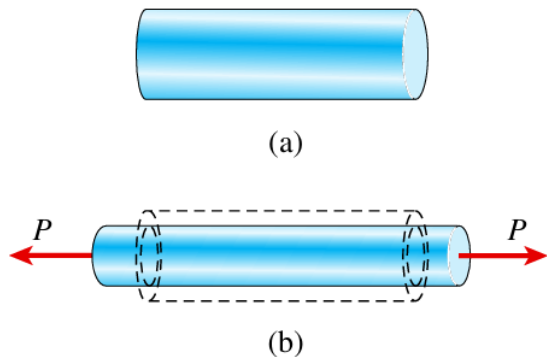
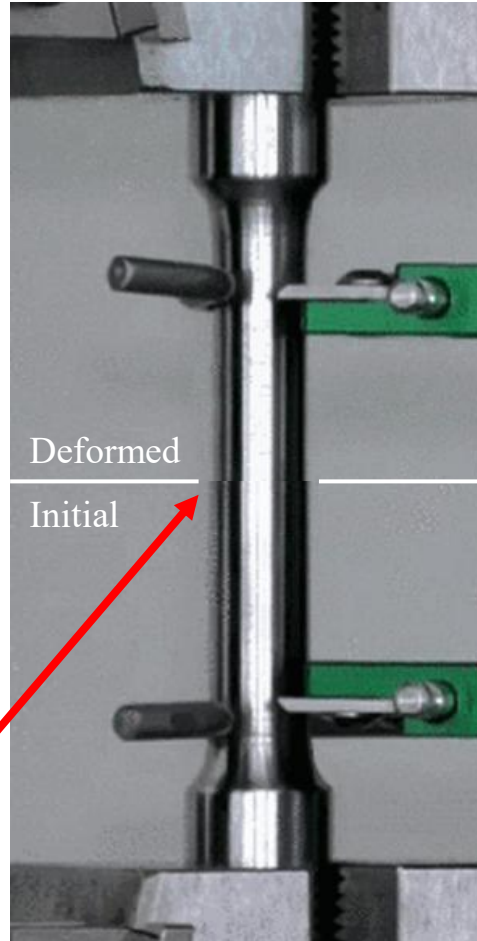
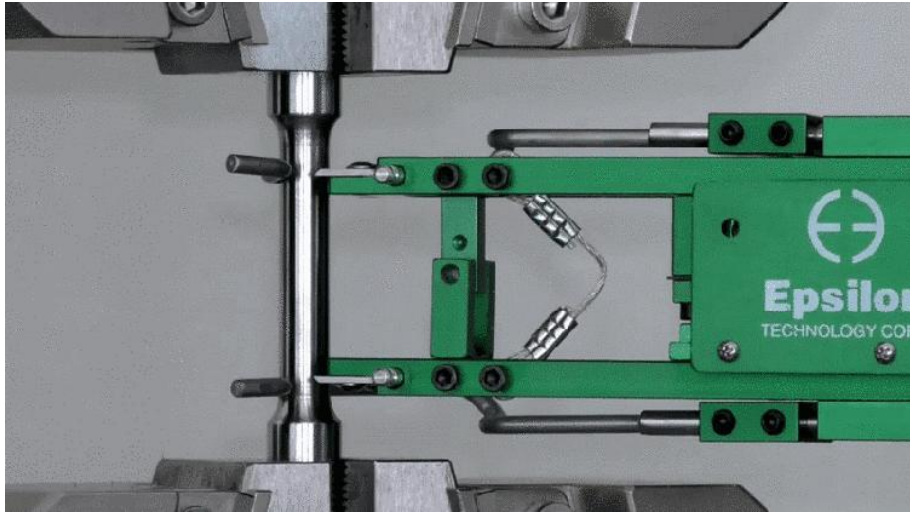
2D assumption

What gives rise to the discrepancy in stress and strain?



	Plane stress	Plane strain
Stresses	$\sigma_z = 0$ $\tau_{xz} = 0$ $\tau_{yz} = 0$ σ_x , σ_y , and τ_{xy} may have nonzero values	$\tau_{xz} = 0$ $\tau_{yz} = 0$ σ_x , σ_y , σ_z , and τ_{xy} may have nonzero values
Strains	$\gamma_{xz} = 0$ $\gamma_{yz} = 0$ ϵ_x , ϵ_y , ϵ_z , and γ_{xy} may have nonzero values	$\epsilon_z = 0$ $\gamma_{xz} = 0$ $\gamma_{yz} = 0$ ϵ_x , ϵ_y , and γ_{xy} may have nonzero values

Revisit Poisson's effect in axial loading



Diameter changes

Poisson's ratio is defined as:

$$\nu = -\frac{\text{lateral strain}}{\text{axial strain}} = -\frac{\epsilon_y}{\epsilon_x} = -\frac{\epsilon_z}{\epsilon_x}$$

Therefore,

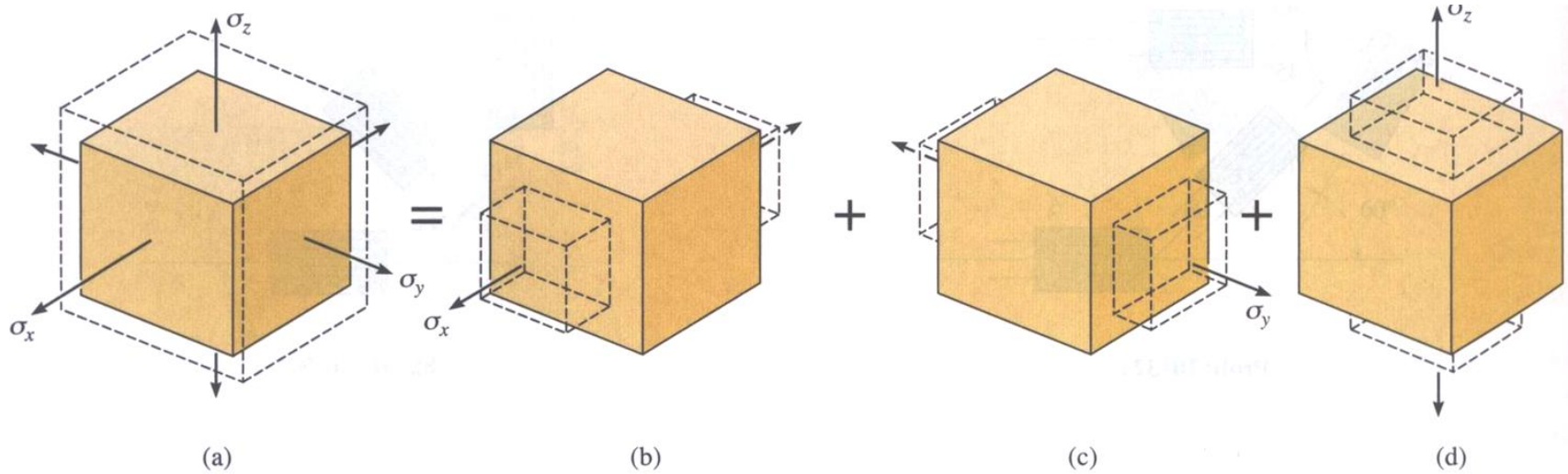
When we have P , thus σ_x

$$\epsilon_x = \frac{\sigma_x}{E}$$

$$\epsilon_y = -\nu\epsilon_x = -\nu\frac{\sigma_x}{E}$$

$$\epsilon_z = -\nu\epsilon_x = -\nu\frac{\sigma_x}{E}$$

3D case: Normal strains



$$\epsilon_x = \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E} - \nu \frac{\sigma_z}{E}$$

$$\epsilon_x = \frac{\sigma_x}{E}$$

$$\epsilon_x = -\nu \frac{\sigma_y}{E}$$

$$\epsilon_x = -\nu \frac{\sigma_z}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \nu \frac{\sigma_x}{E} - \nu \frac{\sigma_z}{E}$$



$$\epsilon_y = -\nu \frac{\sigma_x}{E}$$



$$\epsilon_y = \frac{\sigma_y}{E}$$



$$\epsilon_y = -\nu \frac{\sigma_z}{E}$$

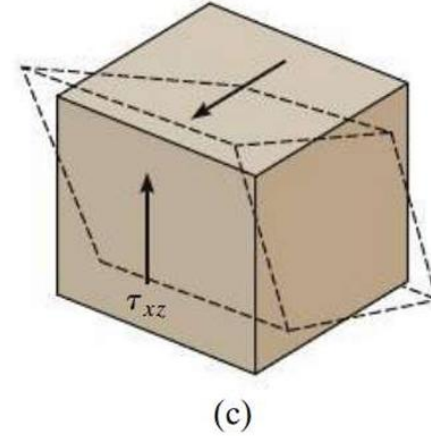
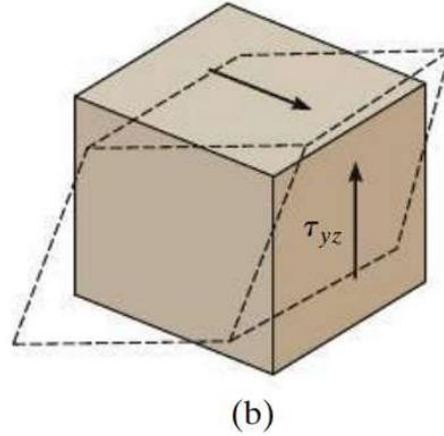
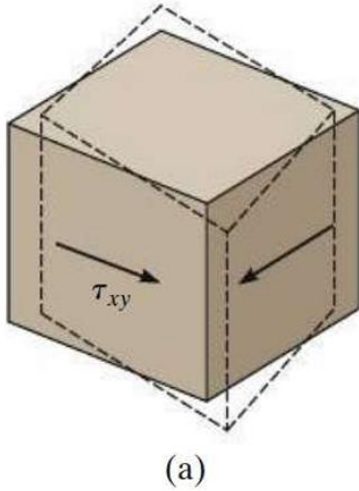
$$\epsilon_z = \frac{\sigma_z}{E} - \nu \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E}$$

$$\epsilon_z = -\nu \frac{\sigma_x}{E}$$

$$\epsilon_z = -\nu \frac{\sigma_y}{E}$$

$$\epsilon_z = \frac{\sigma_z}{E}$$

3D case: Shear strains



If we now apply a shear stress τ_{xy} to the element, experimental observations indicate that the material will deform only due to a shear strain γ_{xy} ; that is, τ_{xy} will not cause other strains in the material.

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$

Generalized Hooke's law

$$\varepsilon_x = \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E} - \nu \frac{\sigma_z}{E}$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \nu \frac{\sigma_x}{E} - \nu \frac{\sigma_z}{E}$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \nu \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E}$$



$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$

Relation between G and E :

$$G = \frac{E}{2(1 + \nu)}$$



$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 + 2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 + 2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 + 2\nu \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix}$$

Example: Hooke's law for plane stress

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

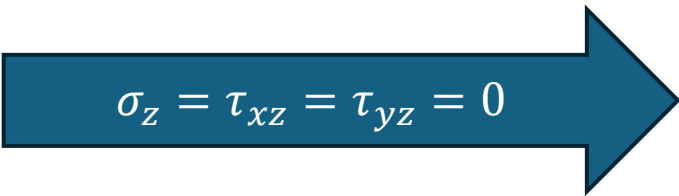
$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$


$$\sigma_z = \tau_{xz} = \tau_{yz} = 0$$

$$\varepsilon_x = \frac{1}{E} (\sigma_x - \nu\sigma_y)$$

$$\varepsilon_y = \frac{1}{E} (\sigma_y - \nu\sigma_x)$$

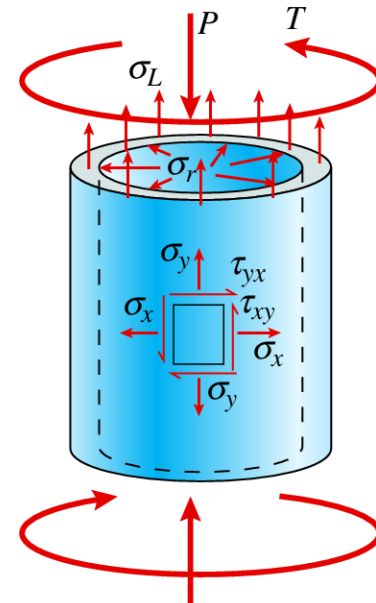
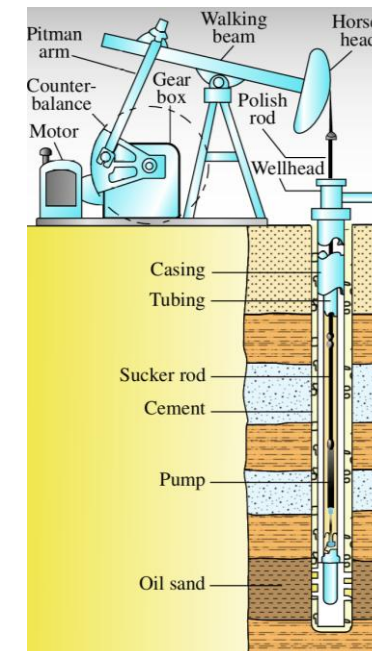
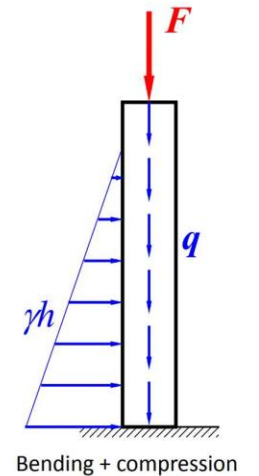
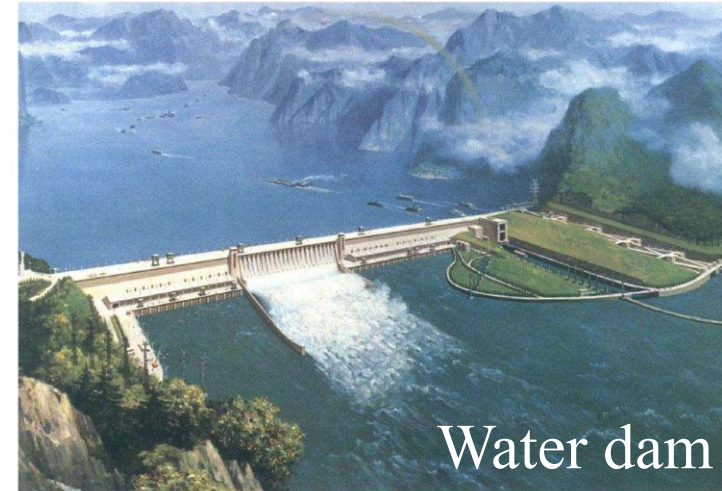
$$\varepsilon_z = -\frac{\nu}{E} (\sigma_x + \sigma_y)$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

Combined load

In engineering practice, a structural member normally exhibits two or more different basic deformations under the action of applied loads.

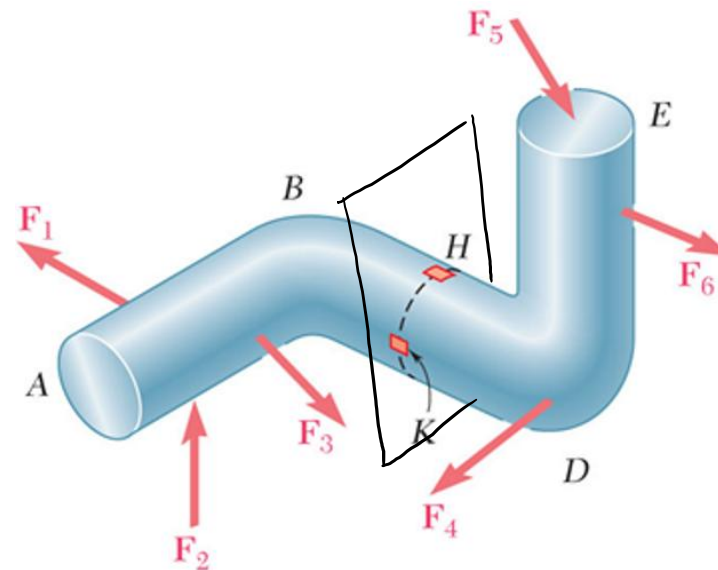
- If **one deformation is primary** while stress (or strain) caused by other deformations is very small, then the case can treat as **basic deformation**.
- In many structures the members are required to resist more than one kind of loading with **almost the same level (combined loading)**.



Method of analysis

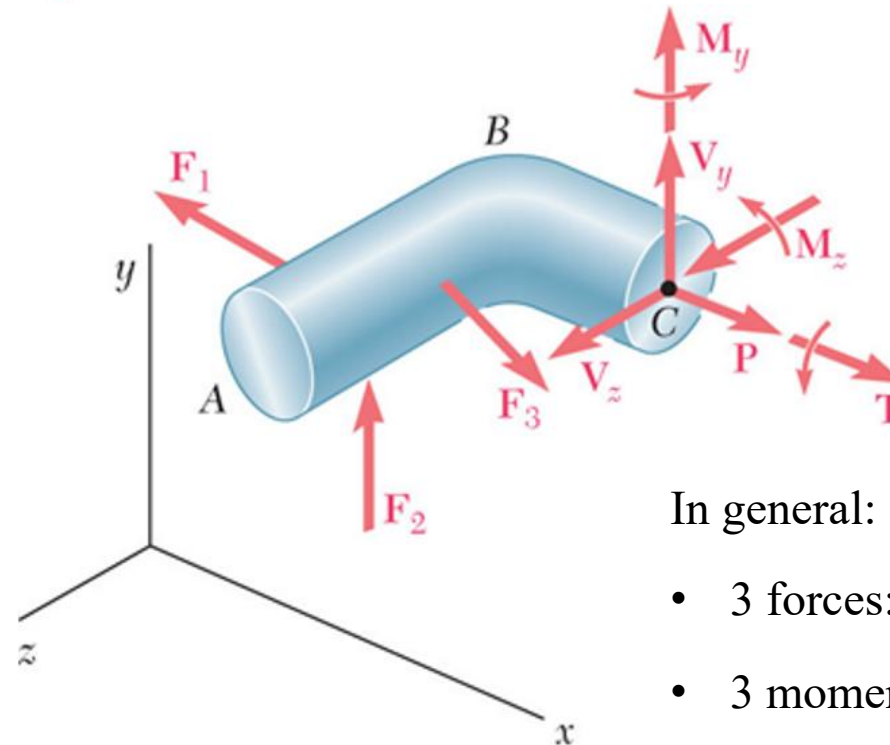
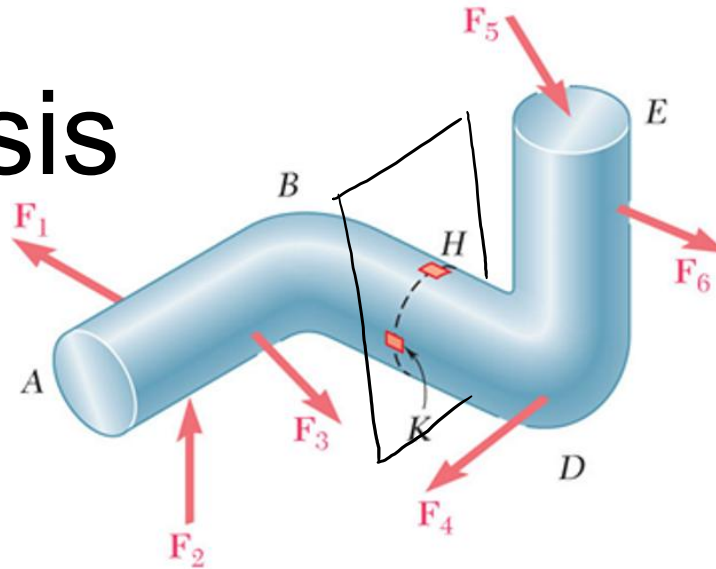
1. Cut through point of interest
2. Expose internal forces and moments
3. Compute stresses corresponding to individual internal forces and moments
4. Combine stresses
5. Get principal / max shear
6. Find strains
7. Repeat at key points

A structural member is subject to arbitrary loadings, we want to know the stress state at K and H



Method of analysis

1. Cut through point of interest
2. Expose internal forces and moments
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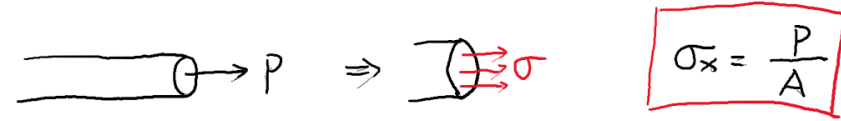
In general:

- 3 forces: P , V_y , V_z
- 3 moments: T , M_z , M_y

Method of analysis

1. Cut through point of interest
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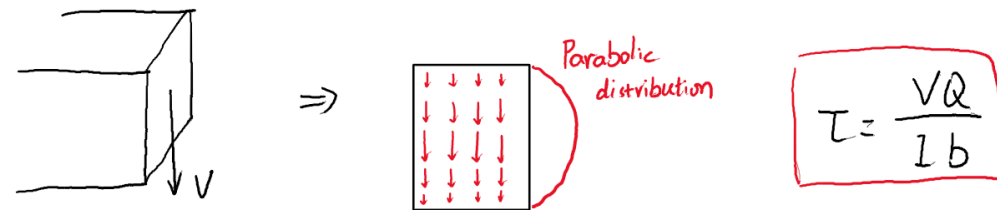
① Normal force P : tension compression.



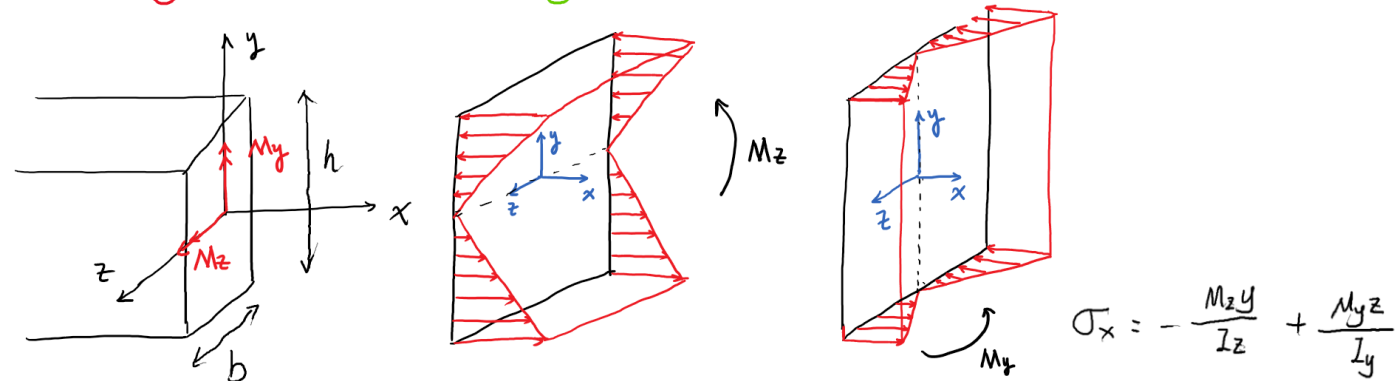
② Torsional Moment T : torsion (circular shaft).



③ Shear force V : bending with shear (Lecture 14)

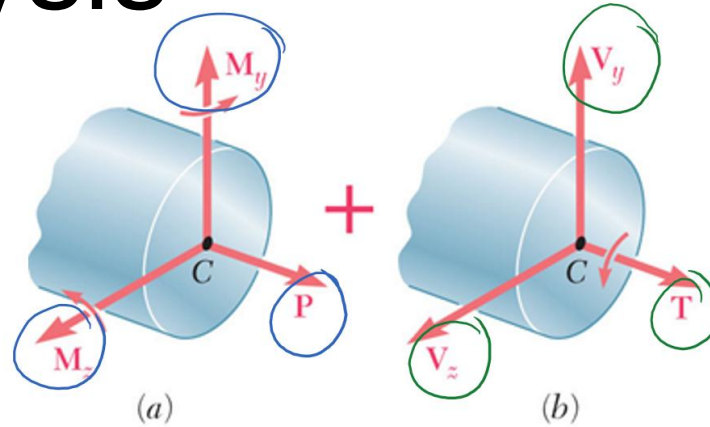


④ Bending moment M : bending (Lecture 12)



Method of analysis

axial force and
in-plane moments
 $\Rightarrow$ normal stress



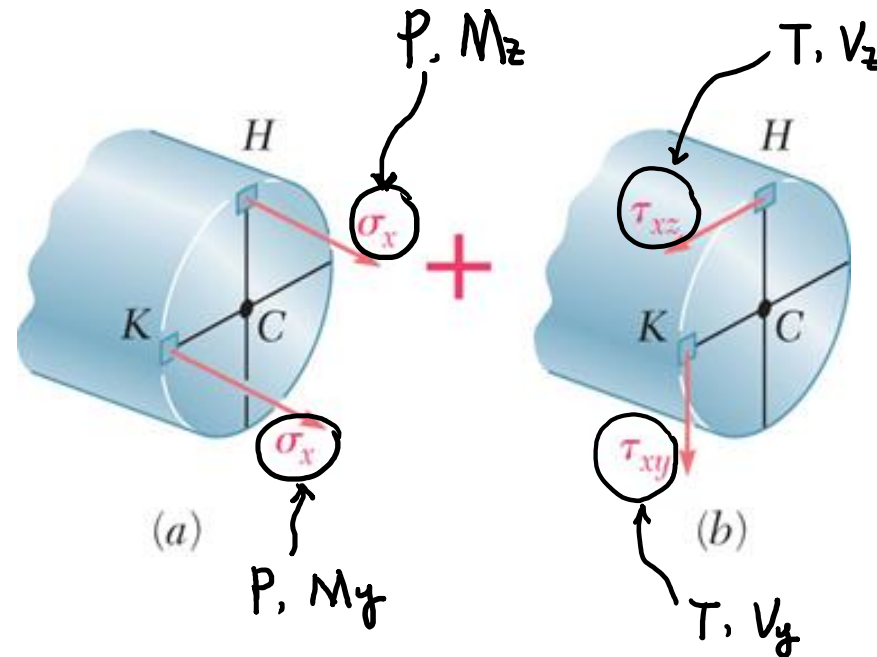
shear force and
axial twisting moment
 $\Rightarrow$ shear stress

Q: at point H, what give me non-zero σ_x ?

- (A) P.
- (B) P. M_y .
- (C) P. M_z
- (D) P. M_y . M_z

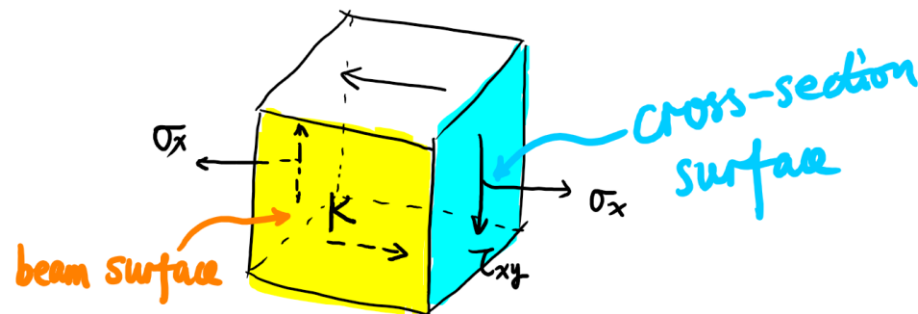
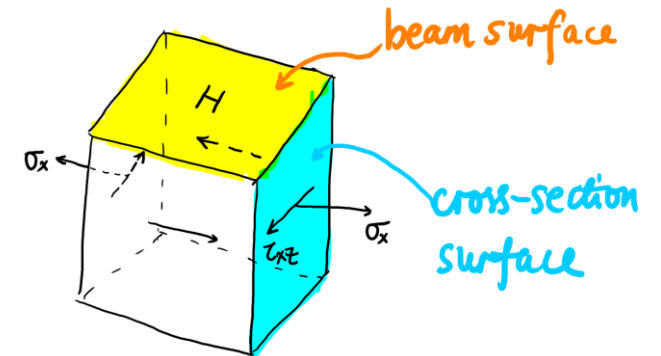
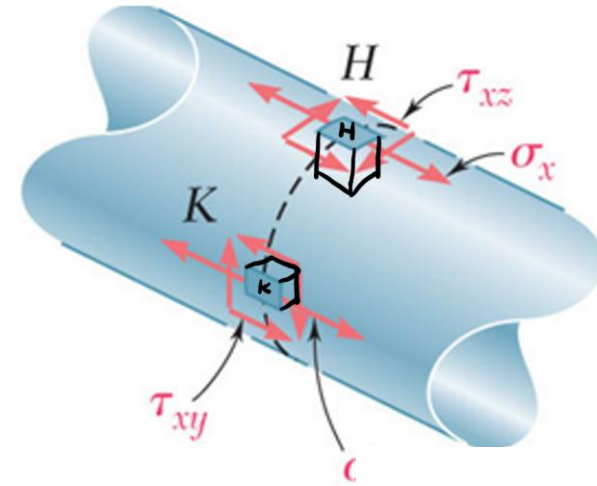
Q: at point H, what give non-zero τ ?

- (A) T.
- (B) T. V_z
- (C) T. V_y
- (D) T. V_z . V_y

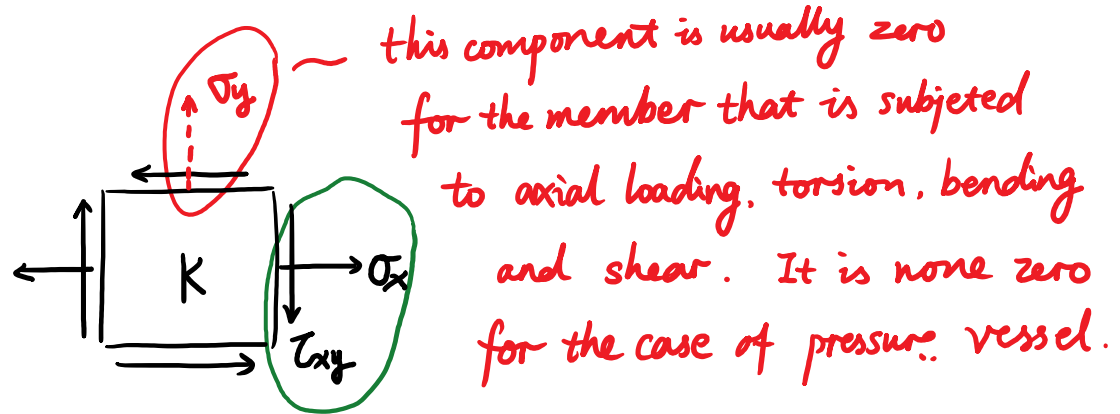


Method of analysis

1. Cut through point of interest
2. Expose internal forces and moments
3. Compute stresses corresponding to individual internal forces and moments
4. Combine stresses
5. Get principal / max shear
6. Find strains
7. Repeat at key points



Method of analysis



These two components are calculated on the crosssection surface, and the stress on the other three surface are subsequently drawn.

